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Tsi12M

Tenkasi District

Common Quarterly Examination - September 2025

15-09-25.

Standard 12

MATHEMATICS

Time Allowed: 3.00 Hours Maximum Marks: 90

I. Choose the best: **20×1=20**

1) If $|\text{adj}(\text{adj } A)| = |A|^n$, then the order of square matrix A is

a) 3 b) 4 c) 2 d) 5

2) If $P = \begin{bmatrix} 1 & x & 0 \\ 1 & 3 & 0 \\ 2 & 4 & -2 \end{bmatrix}$ is the adjoint of 3×3 matrix A and $|A| = 4$, then x is

a) 15 b) 12 c) 14 d) 11

3) The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is

a) 1 b) 2 c) 4 d) 3

4) If z is a non-zero complex number such that $2iz^3 = \bar{z}$ then $|z|$ is

a) $\frac{1}{2}$ b) 1 c) 2 d) 3

5) The solution of the equation $|z| - z = 1 + 2i$ is

a) $\frac{3}{2} - 2i$ b) $-\frac{3}{2} + 2i$ c) $2 - \frac{3}{2}i$ d) $2 + \frac{3}{2}i$

6) If $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is

a) $\frac{1}{2}$ b) 1 c) 2 d) 3

7) If α, β and γ are the zeros of $x^3 + px^2 + qx + \gamma$ then $\sum\frac{1}{\alpha}$ is

a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

8) If $x^3 + 12x^2 + 10ax + 1999$ definetly has a +ve zero, If and only if

a) $a \geq 0$ b) $a > 0$ c) $a < 0$ d) $a \leq 0$

9) If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$ then $\cos^{-1} x + \cos^{-1} y =$

a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π

10) $\sin^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$

a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$

11) The equation $\tan^{-1} x - \cot^{-1} x = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ has

a) no solution b) unique solution

c) two solutions d) infinite number of solutions

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12) If P(x,y) be any point on $16x^2 + 25y^2 = 400$ with Foci F_1 (3,0) and F_2 (−3,0) then $PF_1 + PF_2$ =

a) 8 b) 6 c) 10 d) 12

13) The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is

a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

14) The co-ordinates of one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + C = 0$ are (11,2), the co-ordinates of the other end are

a) (−5,2) b) (−3,2) c) (5,−2) d) (−2,5)

15) If $a = i + j + k$, $b = i + j$, $c = i$ and $(a \times b) \times c = \lambda a + \mu b$ then the value of $\lambda + \mu$ is

a) 0 b) 1 c) 6 d) 3

16) The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is

a) $\frac{\sqrt{7}}{2\sqrt{2}}$ b) $\frac{7}{2}$ c) $\frac{\sqrt{7}}{2}$ d) $\frac{7}{2\sqrt{2}}$

17) The planes $r.(2i - \lambda j + k) = 3$ and $r.(4i + j - \mu k) = 5$ are parallel, then the value of λ and μ are

a) $\frac{1}{2}, -2$ b) $-\frac{1}{2}, 2$ c) $-\frac{1}{2}, -2$ d) $\frac{1}{2}, 2$

18) If A is a square matrix with $|A| = 2$ then for any integer n, the value of $|A^n|$ =

a) 0 b) 2n c) 2^n d) n^2

19) The equation of directrix of the parabola $y^2 + 4y + 4x + 2 = 0$ is

a) $x = -1$ b) $x = 1$ c) $x = -\frac{3}{2}$ d) $x = \frac{3}{2}$

20) If $a = 3i + j - 2k$ and $b = i - 3j + 4k$ are diagonals of a parallelogram then its area is

a) 4 b) $2\sqrt{3}$ c) $4\sqrt{3}$ d) $5\sqrt{3}$

II. Answer any 7 questions. Q.No.30 is compulsory: **7×2=14**

21) Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

22) If adj $A = \begin{bmatrix} 2 & -4 & 2 \\ -3 & 12 & -7 \\ -2 & 0 & 2 \end{bmatrix}$ find A.

23) Which one of the points $10 - 8i$, $11 + 6i$ is closest to $1 + i$?

24) If $z_1 = 6 + 7i$, $z_2 = 3 - 5i$ then find $z_1 + z_2$ and $z_1 z_2$.

25) Construct a cubic equation whose roots are 1, 1 and −2.

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26) Find the principle value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

27) Find the value of $\sec^{-1}\left(\frac{-2\sqrt{3}}{3}\right)$.

28) Find the volume of the parallelopiped whose co-terminus edges are represented by the vectors $-6i+14j+10k$, $14i-10j-6k$ and $2i+4j-2k$.

29) Find the angle between the lines $\frac{x+4}{3}=\frac{y-7}{4}=\frac{z+5}{5}$ and $r = 4k + t(2i+j+k)$.

30) Find the equation of circle for which (2,5) and (−3,8) are the end points of diameter of the circle.

III. Answer any 7 questions. Q.NO:40 is compulsory: **7×3=21**

31) Find the rank of the matrix $\begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$ by reducing it to an echelon form.

32) A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of a 40% acid solution?

33) Find the square root of $6 - 8i$.

34) If $\omega \neq 1$ is a cube root of unity, show that $(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128$.

35) Solve the equation $9x^3 - 36x^2 + 44x - 16 = 0$, if the roots form an arithmetic progression.

36) Find the value of $\tan^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.

37) Find the domain of $\sin^{-1}(2-3x^2)$.

38) If the normal at the point t, on the parabola $y^2 = 4ax$ meets the parabola again the point t_2 then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

39) Find the shortest distance between the two given straight lines $r = (2i+3j+4k) + t(-2i+j-2k)$ and $\frac{x-3}{2}=\frac{y}{-1}=\frac{z+2}{2}$.

40) Find the magnitude and the direction cosines of the torque about the point (1,−2,3) of a force $2i+3j+6k$ whose line of action passes through the origin.

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IV. Answer the following questions: **7×5=35**

41) Find the value k for which the equations $kx - 2y + z = 1$, $x - 2ky + z = -2$, $x - 2y + kz = 1$ have

(i) no solution

(ii) unique solution

(iii) infintely many solution

(OR)

Find all cube roots of $\sqrt{3}+i$.

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42) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$. Show that $x^2 + y^2 = 1$.

(OR)

Find the number of solutions of the equation

$\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$.

43) Solve the equation $(2x - 3)(6x - 1)(3x - 2)(x - 2) - 5 = 0$.

(OR)

Identity the type of conic and find centre, foci, vertices and directrices of $18x^2 + 12y^2 = 144x + 48y + 120 = 0$.

44) By vector method prove that $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$.

(OR)

If z_1, z_2 and z_3 are three complex numbers such that $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$. Show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$.

45) Find the non-parametric of vector equation and cartesian equation of the plane passing through the points (2,2,1), (9,3,6) and perpendicular to the plane $2x + 6y + 6z = 9$.

(OR)

A boy is walking along the path $y = ax^2 + bx + c$ through the points (−6,8), (−2,−12) and (3,8). He wants to meet his friend at P(7,60). Will he meet his friend?

46) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$.

Show that $x^2 + y^2 + z^2 + 2xyz = 1$.

(OR)

If $a = -2i+3j-2k$, $b = 3i-j+3k$, $c = 2i-5j+k$ find $(a\times b)\times c$ and $a\times(b\times c)$ state whether they are equal.

47) A bridge has a parabolic arch that is 10 m high in the centre and 30 m wide at the bottom. Find the height of the arch 6 m from the centre on either sides.

(OR)

Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

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